

Dissecting the Efficacy and Immunogenicity of TLR7 Agonist–Antibody Conjugates through the Lens of Fc Effector Function, Conjugation Strategies, and Linker Cleavability

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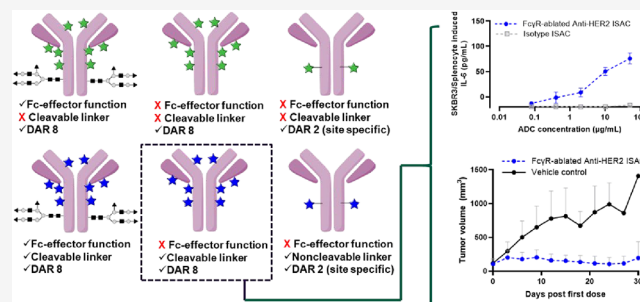
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ABSTRACT: The advent of immuno-oncology therapeutics has given rise to antibody-drug conjugates (ADC) designed to selectively activate pattern recognition receptors such as Toll-Like Receptors (TLRs). In contrast to classical ADC technology, these Immune Stimulating Antibody Conjugates (ISACs) are widely reported to rely on Fc-mediated uptake and noncleavable linkers. Herein, we systematically study the impact of Fc effector function, linker cleavability, and conjugation strategy on the efficacy and immunogenicity of a series of TLR7 agonist-based ISACs. We demonstrate that the combination of FcγR-ablation and incorporation of a cleavable linker results in increased stimulation of myeloid cells, enhanced efficacy, and improved tolerability as compared to traditional ISAC designs. Interestingly, we found that all ISAC designs exhibited evidence of antidrug-antibody (ADA) induction. We found that the combination of a cleavable linker, a highly permeable TLR7 agonist payload, and ablation of FcγR binding results in ISACs that exhibit particularly favorable properties for continued development.



INTRODUCTION

Checkpoint inhibitors have revolutionized cancer treatment by enhancing T-cell responses against tumors, leading to long-term tumor regression. These inhibitors, including anti-PD-1/PD-L1 and anti-CTLA4, are now considered standard treatments for various cancers, including melanoma, nonsmall cell lung cancer, and renal cell carcinoma. However, despite tremendous clinical advances in recent years, the response rate to checkpoint inhibitors remains abysmal. Recent estimates suggest that about 44% of cancer patients are eligible for treatment with immune checkpoint inhibitors (ICI), but only ~12–15% of those patients respond to the treatment.^{1,2} Many cancers, including prostate, breast, and pancreatic, seem inherently resistant to ICI therapy.³ The low response rates and the lack of sensitivity of many cancer types illustrate the urgent need for novel immunotherapy approaches.

One such approach that has been widely embraced is the activation of pattern-recognition receptors (PRRs). Immune checkpoints act as “brakes” that prevent the overactivation and exhaustion of T-cells by MHC-I and MHC-II complexes on cancer cells and antigen-presenting cells (APCs). However, the low response rates mentioned above suggest that simply “releasing the brakes” has only limited therapeutic utility. In contrast, the activation of PRRs represents a plausible strategy to “step on the gas” by turning on various signaling pathways that promote cytokine release, lymphocyte proliferation, and

APC activation. PRRs form a critical link between the innate and adaptive immune systems by recognizing danger-associated molecular patterns (DAMPs). PRRs are highly expressed in many lymphocytes, including those found in “tertiary lymphoid structures” within the tumor microenvironment.⁴ Tertiary lymphoid structures (TLS) share many features with secondary lymphoid organs, and their presence in tumors is associated with favorable clinical outcomes in multiple types of cancer.⁴ Activation of various PRRs is associated with the formation of TLS, suggestive of therapeutic utility.^{5,6}

One PRR of particular interest is Toll-Like Receptor 7 (TLR7). TLR7 is an endosomal PRR that recognizes single-stranded RNA and plays an important role in antiviral responses. A modestly potent TLR7 agonist, Imiquimod, received FDA approval as a topical antiviral agent in 1997 and went on to be approved for the treatment of basal cell carcinoma in 2003.^{7,8} Recently, topical administration of this

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compound has been shown to be effective in the treatment of melanoma and breast cancer skin metastasis. Its activity appears to be driven by the generation of various pro-inflammatory cytokines, including IFN- α , TNF- α , IL-1, IL-6, and IL-8, and also by its ability to recruit and activate dendritic cells.⁹ Despite these promising results as a topical agent, systemic administration of Imiquimod has been associated with severe immunological adverse events, thus restricting its application to topical administration for skin-associated cancers.¹⁰ For this reason, there has been a flurry of technologies recently explored for tumor-specific delivery of TLR agonists, including intratumoral injections, nanoparticle formulations, prodrugs cleaved in the tumor microenvironment, and (most recently) antibody-conjugates.^{11–14}

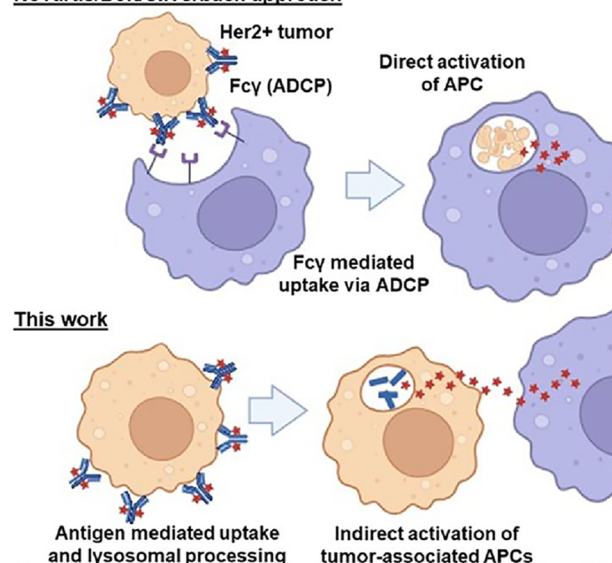
Recent advancements in antibody-drug-conjugate (ADC) technology have enabled the delivery of highly toxic payloads to tumor tissue while minimizing systemic exposure.¹⁵ There are now 12 FDA-approved ADCs that are marketed in the United States, 7 of which are used to treat solid tumors. ADCs generally consist of three components: (1) An antibody that binds to a tumor-selective antigen; (2) A cytotoxic payload; and (3) A linker connecting these two moieties that is cleaved upon uptake in tumor tissue. There have been numerous technical advances in recent years that have reduced the systemic (off-tumor) release of payload, thus minimizing the risks associated with systemic exposure. In light of the success in delivering cytotoxic payloads, it is not surprising that various teams are now exploring the delivery of immune-stimulating payloads using ADC technology. These so-called Immune Stimulating Antibody Conjugates (ISACs) have been employed for the delivery of various PRR agonists, including TLR7, TLR8, and STING agonists.

ISAC approaches employing TLR agonists have rapidly advanced into the clinic, but have run into significant hurdles.^{16,17} For example, NJH395 (Novartis) was evaluated in 14 patients—and antidrug-antibodies (ADA) were detected in all patients.¹⁷ These immunogenic concerns, combined with only modest efficacy signals at the tolerated dose, resulted in the abandonment of this program. A related program from Bolt did not report ADA, but rather faced very significant PK problems that severely impeded drug efficacy and resulted in the termination of the clinical trial.¹⁸ Likewise, a TLR8 agonist ISAC program from Silverback resulted in significant immunological toxicities that resulted in termination of the program.¹⁹

The failure of these programs illustrates the need for new approaches for ISAC designs. Unlike cytotoxic ADCs, most current generation ISACs are reported to primarily rely on antibody-directed cell-mediated phagocytosis (ADCP). As illustrated in Figure 1, ADCP is driven by Fc- γ receptor activation on antigen-presenting cells, particularly tumor-associated macrophages (TAMs). Upon phagocytosis, the tumor cell and the associated ISAC are degraded in the lysosome, resulting in the release of the TLR7 agonist. Importantly, Fc- γ mediated uptake may occur in nontumor tissues and is speculated to be a driver of various immunotoxicities for traditional ADC technology.^{20,21} Moreover, the use of noncleavable linkers results in a cysteine-linked payload that is generally believed to have low permeability, thus allowing the buildup of TLR7 agonists even in situations where the uptake rate is low.

For these reasons, our team has been investigating new ISAC designs that employ cleavable linkers attached to highly

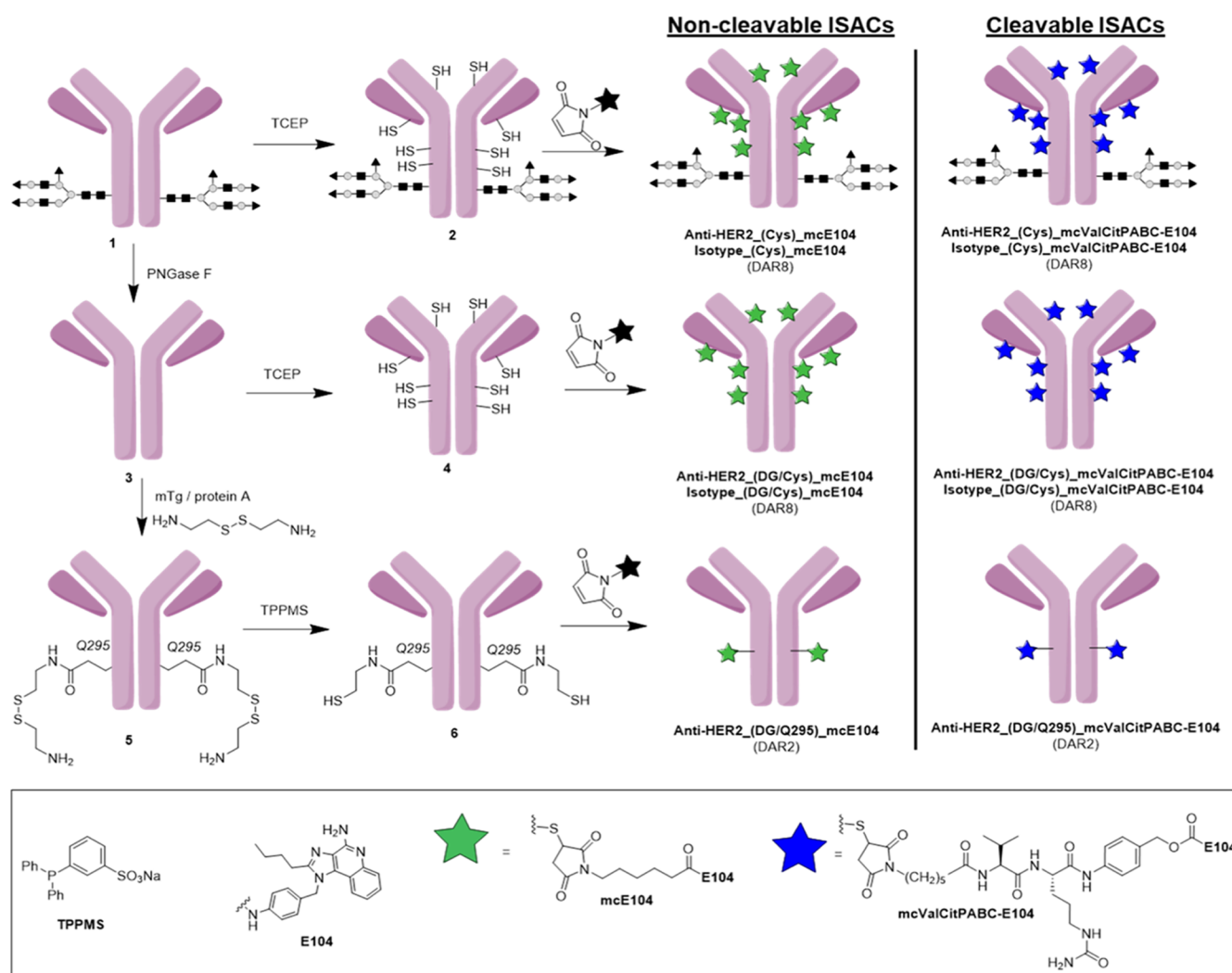
Novartis/Bolt/Silverback approach



ADC / Company	Antibody target	Payload class	Linker type	Fc γ ?	Status?
BDC-1001 Bolt Bio.	HER2	TLR7/8	Non-cleavable	Yes	Phase 2 Discontinued
NJH-395 Novartis	HER2	TLR7	Non-cleavable	Yes	Phase 1 Discontinued
SBT-6050 Silverback	HER2	TLR8	Non-cleavable (?)	Yes	Phase 1 Discontinued
Tumey Mol. Pharm. 2022	HER2	TLR7	Cleavable	No	Preclinical

Figure 1. Mechanism of action of immune-stimulating ADCs (top) and summary of discontinued immune-stimulating ADC programs.

permeable TLR7 agonists. Importantly, this new design does not rely on Fc- γ receptor (Fc γ R) binding, thus allowing us to employ Fc- γ ablated antibody backbones. In 2022, we reported the first generation of this new design, clearly showing that the redesigned ISACs exhibit comparable efficacy to the traditional ISAC designs.²² However, our first disclosure of this work did not systematically explore how Fc γ R binding affinity and the linker type impacted efficacy, immunogenicity, and PK. Herein, we report the design and preparation of a set of ISACs in which we systematically vary the linker type, Fc γ R binding affinity, and conjugation technology. We show that while all designs retain a degree of efficacy in coculture models and a tumor xenograft model, ablation of the Fc γ R binding affinity results in increased efficacy, potency, and tolerability. Interestingly, we found that all ISAC designs resulted in rapid clearance after the second dose to immunologically competent mice, suggestive of ADA formation. The formation of ADA beginning ~96 h after the second dose, was confirmed with a ligand-binding assay (LBA). Importantly, the formation of ADA was associated with an increase in adverse events, particularly for the noncleavable linkers. Finally, we show that site-specific conjugation at the Q295 site may impart a degree of protection against immunogenicity. Taken together, our results provide strong evidence that the combination of a cleavable linker, a highly permeable TLR7 agonist payload, and ablation of Fc γ R binding results in ISACs that exhibit favorable properties for continued development.

Scheme 1. Conjugation Chemistry Used to Prepare Endogenous Cysteine Conjugates and Site-Specific Q295 Conjugates⁴²

⁴²Note that the non-cleavable mcE104 is denoted in green and the cleavable mcValCitPABC E104 is denoted in blue throughout this document.

RESULTS

Design and Preparation of Bioconjugates. In 2022, our team reported the identification of a potent imidazo[4,5-*c*]quinoline TLR7 agonist referred to as “E104”.^{22,23} This compound is a potent and selective agonist of both mouse and human TLR7, inducing the activation of the NFκB pathway and cytokine release in TLR7-expressing lymphocytes and myeloid cells at low nM concentrations. Further, we demonstrated that this compound induced upregulation of T-cell coreceptors (CD80/86) in PBMCs and was highly selective (>1000×) for TLR7 over TLR8. One important feature of this compound is its high permeability—an essential attribute in the design of “bystander active” ADC payloads. Our prior work demonstrated that tumor-targeting bioconjugates of E104 employing a cleavable linker exhibit potent antitumor activity, presumably by a “bystander” type of mechanism. However, our prior work did not explicitly assess the Fcγ-binding affinity of these conjugates or directly compare the activity of cleavable and noncleavable linkers.

With this in mind, we prepared a series of ISACs that employed either a noncleavable (mcE104) or cleavable (mcValCitPABC_E104) linker. (Scheme 1) These linker-payloads were attached to a HER2-targeting antibody

(trastuzumab) or to an appropriate nonbinding isotype control (palivizumab or rituximab). In order to understand the role of FcγR-binding, conjugates were prepared from both glycosylated (FcγR active) and deglycosylated (FcγR ablated) antibodies (1 and 3, respectively). As will be extensively discussed below, removal of the N297 glycosyl group with the endoglycosidase PNGaseF is a widely employed method for ablation of FcγR binding affinity.²⁴ Reduction of the endogenous intrachain disulfides with TCEP gave intermediates 2 and 4, which was followed by the addition of the appropriate maleimide-containing linker-payload, as illustrated in Scheme 1. In order to maximize payload delivery and minimize product heterogeneity, we employed a “full reduction” with excess TCEP (~6 equiv) followed by an excess (~10 equiv) of the linker-payload. This generally resulting a homogeneous drug-to-antibody-ratio (DAR) of ~8. As shown in Table 1, aggregation was generally low (<5%) and the observed mass shift of the heavy chain and light chain was within 3–4 Da of the expected shift (generally within the margin of error for electrospray ionization protein MS).

Given that high loading (DAR8) often imparts biophysical problems that lead to aggregation and rapid clearance, we additionally prepared site-specific DAR2 conjugates using our

Table 1. Summary and Characterization of ISACs Prepared^a

ADC Name	Antibody	Conjugation	Glycosylated?	DAR	Aggregation	HIC shift (min)	Theoretical mass shift	Actual mass shift
Anti-HER2_(Cys)_mcE104	Trastuzumab	Cysteine	Yes	7.7	<5%	+4.32	538	539
Anti-HER2_(DG/Cys)_mcE104	Trastuzumab	Cysteine	No	7.8	<5%	+2.46	538	539
Anti-HER2_(DG/Q295)_mcE104	Trastuzumab	Q295-thiol	No	1.9	<3%	+1.29	538	539
Isotype_(Cys)_mcE104	Palivizumab	Cysteine	Yes	8.0	<5%	-	538	541
Isotype_(Cys)_mcE104	Rituximab	Cysteine	Yes	8.0	<5%	-	538	540
Isotype_(DG/Cys)_mcE104	Palivizumab	Cysteine	No	8.0	<5%	-	538	540
Anti-HER2_(Cys)_mcValCitPABC-E104	Trastuzumab	Cysteine	Yes	6.8	6.2%	+3.99	943	945
Anti-HER2_(DG/Cys)_mcValCitPABC-E104	Trastuzumab	Cysteine	No	8.7	11.7%	+4.49	943	944
Anti-HER2_(DG/Q295)_mcValCitPABC-E104	Trastuzumab	Q295-thiol	No	1.8	<3%	+1.96	943	944
Isotype_(Cys)_mcValCitPABC-E104	Palivizumab	Cysteine	Yes	8.0	<5%	-	943	944
Isotype_(DG/Cys)_mcValCitPABC-E104	Palivizumab	Cysteine	No	8.0	<5%	-	943	943
Isotype_(DG/Cys)_mcValCitPABC-E104	Rituximab	Cysteine	No	8.0	6.9%	-	943	943

^aNon-cleavable ISACs are shown in green and cleavable ISACs are shown in blue. The annotation in parentheses indicates the site of conjugation (“Cys” or “Q295”) while the annotation (“DG”) indicates that the antibody was deglycosylated prior to conjugation. Note that anti-RSV (palivizumab) and anti-hCD20 (rituximab) were employed for isotype controls. HIC shift refers to the retention time shift as compared to the unmodified antibody by hydrophobic interaction chromatography.

published transglutaminase-mediated Q295 thiolation chemistry.^{25,26} This method relies on a transamidation reaction that is highly selective for the Q295 residue of aglycosyl antibodies. In short, the N297 glycosyl group is removed by PNGaseF, and the resulting antibody (3) is subjected to a transglutaminase-mediated addition of cystamine to the newly exposed Q295 residue, providing the disulfide-capped intermediate 4. Selective reduction of the newly introduced disulfide is accomplished with triphenylphosphine-*meta*-sulfonic acid (TPPMS) to provide 6. Finally, addition of the appropriate maleimide-containing linker-payload provided ISACs with a DAR of ~2. (Table 1) As anticipated, little or no aggregation was found, and the observed mass shift was in line with the expected mass shift. Of note, the hydrophobic interaction chromatography (HIC) shift for the Q295 conjugates was significantly lower than the matched DAR8 cysteine conjugate, indicative of reduced hydrophobicity.

Comparison of Industry Standard ISACs to the FcγR-Ablated Cleavable-Linker ISACs. At the outset of these studies, it was important to compare our newly reported technology with “industry standard” ISAC technology. To date, there have been two reports of TLR7-agonist ISACs that have entered clinical development (BDC-1001 from Bolt Biotherapeutics and NJH-395 from Novartis), both of which have been discontinued.^{17,18} Both of these ISACs employ fully glycosylated antibodies (*i.e.*, FcγR intact) and noncleavable linkers to the payload. Indeed, prior reports suggest that these features are important for the efficacy of the ISAC. Thus, our initial studies focused on directly comparing the industry standard “NJH395-like” glycosylated/noncleavable ISAC anti-HER2_(Cys)_mc104 with the deglycosylated/cleavable anti-HER2_(DG/Cys)_mcValCitPABC_E104 from our prior work.²²

Importantly, TLR7 has little or no expression in most tumor tissues—and thus, not surprisingly, anti-HER2_(Cys)_mc104 or anti-HER2_(DG/Cys)_mcValCitPABC_E104 exhibited no cytotoxicity against HER2-positive SKBR3 and HCC1974

cells at concentrations up to 30 μg/mL. (Supporting Figure S1) Therefore, we employed a set of coculture assays to assess the potency and mechanism of the two ISACs. Conceptually, the ISAC will bind to the HER2-positive tumor cell while the payload will act on the tumor-associated (cocultured) myeloid or lymphoid cells. In one such permutation, HER2-positive breast cancer cells (SKBR3) were cocultured with TLR7-expressing mouse macrophage cells (RAW264.7) engineered with an NFκB reporter enzyme (secreted embryonic alkaline phosphatase/SEAP). As shown in Figure 2A, the HER2-targeted “industry standard” ISAC (anti-HER2_(Cys)_mcE104) induced activation of the macrophage NFκB pathway at ~5× lower concentrations than the matched isotype control (~1.2 vs ~6 μg/mL). In contrast, our redesigned ISAC (anti-HER2_(DG/Cys)_mcValCitPABC-E104) induced macrophage activation at far lower concentrations (<0.24 μg/mL). Moreover, the matched isotype control of the cleavable ISAC did not elicit any significant macrophage activation until ~30 μg/mL, providing a >100× window between the targeted and nontargeted ISAC. (Figure 2A) Building on this encouraging data, we proceeded to perform a coculture of HCC1954 (HER2+) cells and human PBMCs using the same set of ISACs. As shown in Figure 2B, both the industry standard ISAC (anti-HER2_(Cys)_mc104) and our newly designed bioconjugate (anti-HER2_(DG/Cys)_mcValCitPABC_E104) elicited a robust release of IFNα at ~1–6 μg/mL concentrations, while the matched isotype comparators had a negligible effect at concentrations up to 30 μg/mL. Interestingly, incubation of the noncleavable ISAC (anti-HER2_(Cys)_mcE104) with human PBMCs in monoculture (in the absence of HER2+ cells) resulted in the robust release of TNFα and IL-6 while incubation with the redesigned ISAC (anti-HER2_(DG/Cys)_mcValCitPABC_E104) did not. (Supporting Figure S2) This results likely reflects decreased FcγR-mediated uptake (*vide infra*) and is suggestive of reduced off-target immune-toxicity.

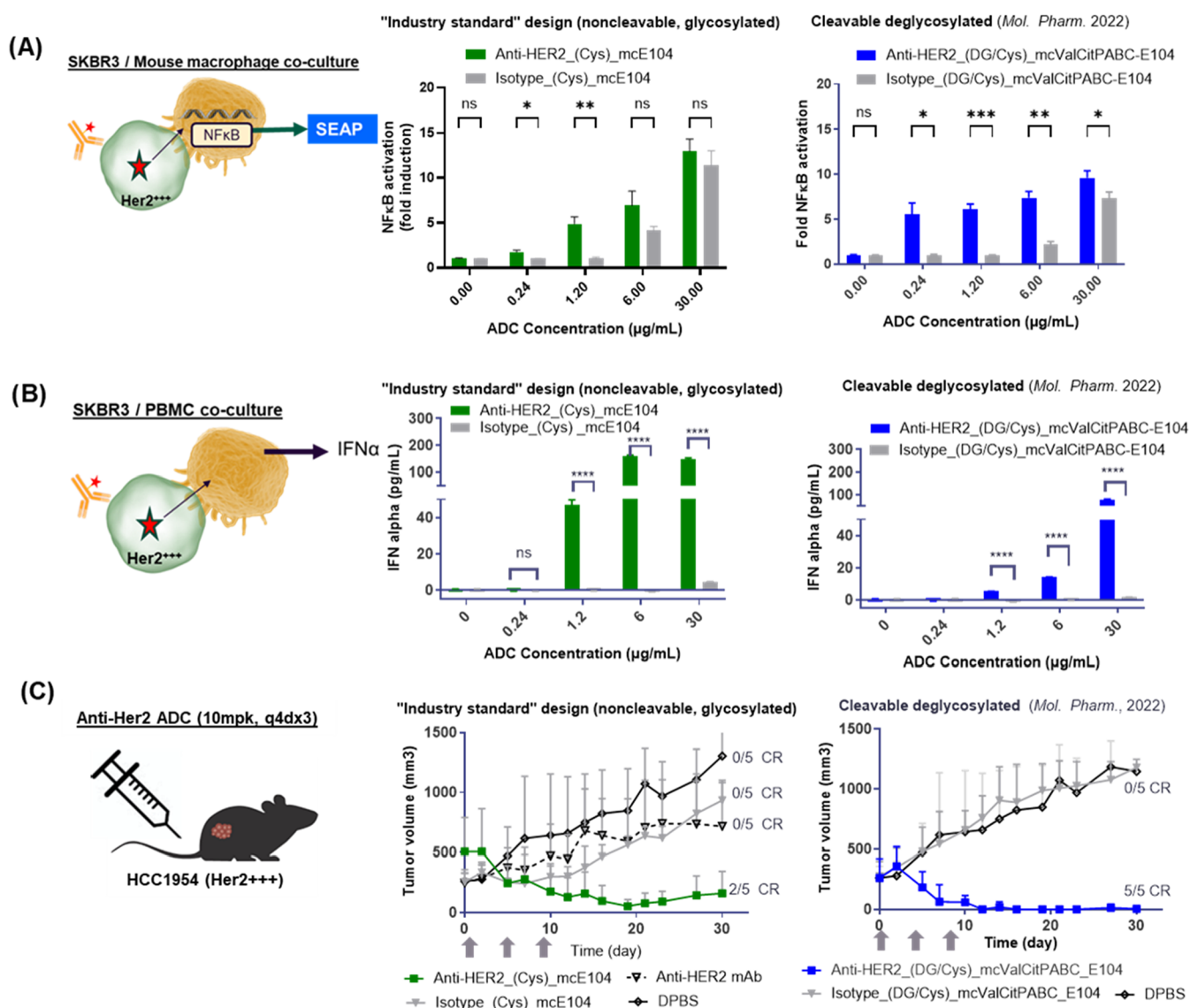


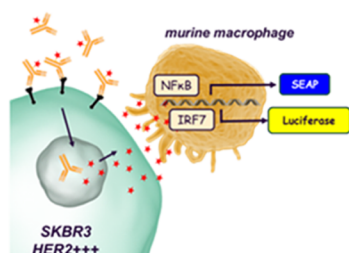
Figure 2. "Industry standard" ISAC design (Anti-HER2_(Cys)_mcE104, left graphs) was compared directly with the redesigned cleavable/deglycosylated ISAC (anti-HER2_(DG/Cys)_mcValCitPABC-E104, right). (A) HER2+ breast cancer cells (SKBR3) were cocultured for 48 h with mouse macrophage reporter cells, looking for activation of the NFκB pathway; (B) HER2+ breast cancer cells (HCC1954) were cocultured with human PBMCs for 24 h, looking for the release of IFNα; and (C) HER2+ breast cancer cells (HCC1954) were implanted in SCID beige mice (n = 5/group), looking for a reduction in tumor volume (IP administration, q5dx3). In each case the HER2 targeted ADC (in green) was compared head-to-head with an isotype (anti-CD20/nontargeted) control ADC (in gray). Note that the right-hand panels of Figures 2A–C represent data previously published by our team, included here for reference.

Having demonstrated that both conjugates exhibited robust and antigen-dependent *in vitro* activity in the presence of HER2+ tumors, we proceeded to evaluate the *in vivo* efficacy in a HER2+ HCC1954 xenograft study using SCID beige mice. While SCID mice lack B and T cells repertoires necessary for adaptive immunity, they retain much of their innate immune system, including myeloid cells. Notably, several studies have shown that TLR-targeting ISACs are active in immune-compromised mouse models.^{16,17,22} As shown in Figure 2C, HCC1954 tumors were grown to ~200–300 mm³ prior to dosing the appropriate ISAC at 10 mg/kg (q5dx3, IP). Consistent with literature reports of similar ISACs, the industry standard "NJH395-like" anti-HER2_(Cys)_mcE104 induced significant tumor regression, ultimately resulting in 2/5 complete responses (CR). As anticipated, the isotype control, naked mAb, and vehicle-treated groups experienced

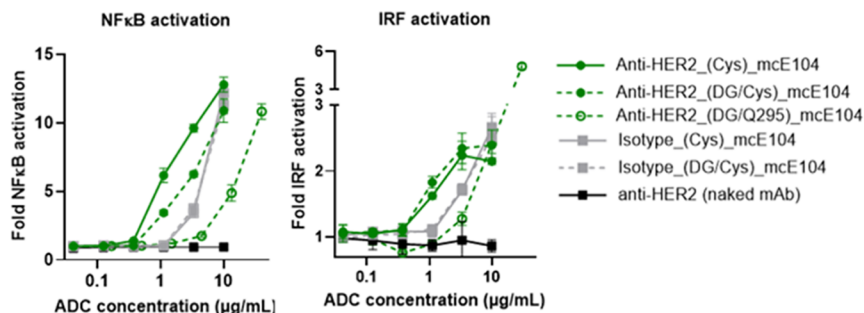
rapid tumor growth and promoted no CR. Excitingly, our redesigned ISAC, anti-HER2_(DG/Cys)_mcValCitPABC-E104, promoted rapid tumor regression with 5/5 CR.²² In contrast, the matched isotype control exhibited no CR, mirroring the vehicle treatment group (Figure 2C).

Impact of Glycosylation, Linker, and Conjugation Technology on ISAC Activity. Having established that our redesigned ISAC (anti-HER2_(DG/Cys)_mcValCitPABC-E104) performs comparably or better than the standard ISAC design, Anti-HER2_(Cys)_mcE104, in a variety of *in vitro* and *in vivo* settings, we proceeded to test the various permutations described above (Scheme 1 and Table 1) in order to understand the impact of glycosylation, linker-type, and conjugation technology on the activity. Once again, our initial studies involved a coculture of HER2+ SKBR3 cells with the murine macrophage RAW264.7 engineered with both an

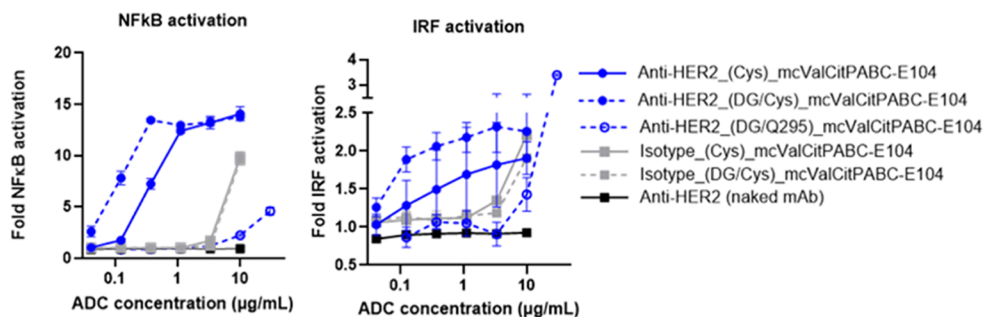
(A) SKBR3 / macrophage co-culture



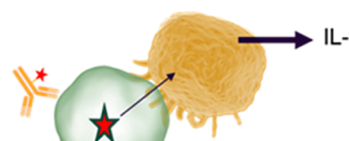
Noncleavable linkers



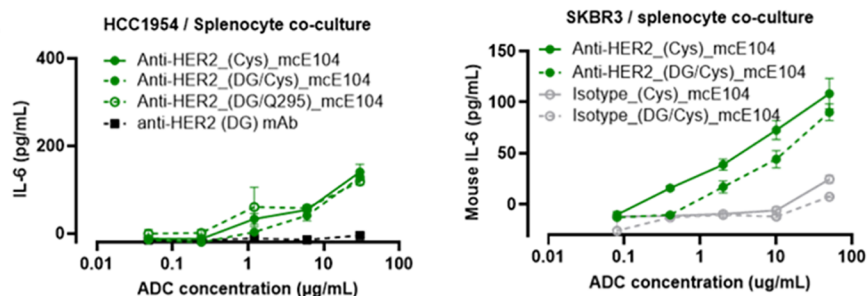
Cleavable linkers



(B) HER2+ cancer / splenocyte co-culture



Noncleavable linkers



Cleavable linkers

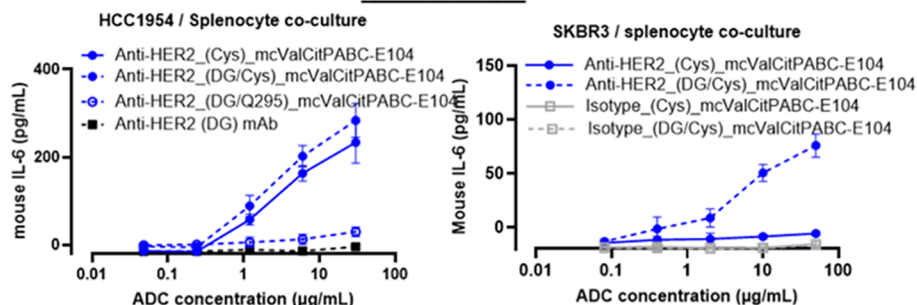


Figure 3. Various permutations of targeted and nontargeted immune-stimulating ADCs were evaluated in tumor/macrophage cocultures (A) and tumor/splenocyte cocultures (B) for 48 and 24 h, respectively. Dotted lines indicated deglycosylated ADCs while solid lines are glycosylated ADCs. Blue/green lines are cleavable (ValCitPABC) and noncleavable (mc) linkers, respectively. Gray lines are isotype (nontargeted) ADC controls generated using an anti-RSV IgG1 (palivizumab). Black lines indicate naked mAb controls.

NFκB and IRF reporter gene (RAW-Dual, Invivogen). The NFκB pathway was linked to a SEAP reporter that could be monitored by UV absorption, while the IRF (Interferon regulatory factors) pathway was linked to a luciferase reporter that could be independently assessed by luminescence. Consistent with the data mentioned above, **anti-HER2_(cys)_mcE104** promotes macrophage activation of both the NFκB and IRF pathways at $\sim 1 \mu\text{g/mL}$, while the matched isotype

control is $\sim 5\times$ less potent. (Figure 3A) Removal of the glycosyl moiety (**anti-HER2_(DG/cys)_mcE104**) has a negligible impact on the activity. The Q295 construct is $\sim 5\times$ less potent than the deglycosylated cysteine conjugate, consistent with the decreased DAR (2 vs 8). It should be noted that the naked anti-HER2 mAb, unsurprisingly, did not elicit any macrophage activation, even at the highest concentrations. Turning to the cleavable linker, we once again observed that

Table 2. Binding to Several Fc γ Isoforms was Evaluated by SPR^a

ADC	Fc- γ RI (K _D , nM)	Fc- γ RIla (K _D , nM)	Fc- γ RIlb (K _D , nM)	Fc- γ RIla (K _D , nM)
Anti-HER2 mAb	12.7	377	2720	119
Anti-HER2 (Cys)_mcE104	9.6	486	3490	108
Anti-HER2 (DG/Cys)_mcE104	>300	>8000	>8000	>2000
Anti-HER2 (DG/Q295)_mcE104	>300	>8000	>8000	>2000
Anti-HER2 (Cys)_mcValCitPABC-E104	5.7	479	1890	123
Anti-HER2 (DG/Cys)_mcValCitPABC-E104	>300	>8000	>8000	>2000
Anti-HER2 (DG/Q295)_mcValCitPABC-E104	>300	>8000	>8000	>2000

^aThe response (total binding, RU) was normalized to the response of the naked mAb. Raw results can be found in the [Supporting Figures](#).

the deglycosylated anti-HER2 (DG/Cys)_mcValCitPABC-E104 promoted macrophage activation (as assessed by both NF κ B and IRF) at concentrations as low as 0.1 μ g/mL while the matched isotype control required $\sim$ 100-fold higher concentrations to elicit activation. (Figure 3A) Interestingly, the matched glycosylated ISAC (anti-HER2 (Cys)_mcValCitPABC-E104) exhibited modestly reduced potency. Quite unexpectedly, the Q295 conjugate displayed little or no activity at concentrations as high as 30 μ g/mL.

Additional coculture experiments were performed using mouse splenocytes and HER2+ SKBR3 or HCC1954 cells. Splenocytes are a common source of mononuclear antigen-presenting cells (APCs). Following the protocol described by Brant, the HER2+ cells were seeded in a 1:10 ratio with freshly thawed splenocytes.²⁷ The activation of splenocytes was assessed by ELISA quantitation of mouse IL-6 in the supernatant after 24 h of incubation. Unlike the macrophage coculture results, we found that the cleavable and noncleavable permutations of the ISACs elicited similar activity, requiring $\sim$ 1–10 μ g/mL of the HER2-targeted ADC to promote release of IL-6. Once again, the naked mAb and isotype controls elicited little or no response, suggestive of antigen-dependent activity. In most cases, the presence or absence of glycosylation had little impact on the activity. However, it is interesting to note that the deglycosylated anti-HER2 (DG/Cys)_mcValCitPABC-E104 elicited significantly more IL-6 release than its glycosylated counterpart in the SKBR3/splenocyte coculture, a phenomenon not observed in the HCC1954/splenocyte coculture. Consistent with the macrophage coculture results, the cleavable Q295 conjugate anti-HER2 (DG/Q295)_mcValCitPABC-E104 was once again virtually inactive.

Impact of Glycosylation and Conjugation on Fc γ R Binding and Effector Function. As mentioned above, the efficacy of nearly all ISACs developed to date are reported to rely on Fc γ R-mediated uptake of opsonized target cells, thus driving payload release and activation of myeloid cells and APCs.^{16,17,27} Therefore, it was imperative that we assess the Fc γ R binding affinity of our panel of ISACs. Humans are known to express six subclasses of Fc γ Rs: Fc γ RIa (CD64a), Fc γ RIIa (CD32a), Fc γ RIIb (CD32b), Fc γ RIIc (CD32c), Fc γ RIIIa (CD16a), and Fc γ RIIIb (CD16b).²⁸ Among these 6 subclasses, five of them are activating Fc receptors, while Fc γ RIIb is a suppressive receptor. The five activating subclasses are predominantly expressed on myeloid cells, such as monocytes, macrophages, and dendritic cells, while the suppressive subtype (Fc γ RIIb) is primarily expressed on B cells. Similar to humans, mice possess three activating Fc γ Rs, namely mouse Fc γ RI (mFc γ RI), mFc γ RIII, and mFc γ RIV, in addition to one inhibitory Fc γ R, mFc γ RIIb. Human IgG1

generally exhibits comparable affinities for human and mouse Fc γ Rs.^{29,30} Therefore, in most cases, human IgG's will induce effector function (ADCP/ADCC) in mouse model systems.

Interestingly, there have been few reports of systematically studying the effect of various conjugation technologies and linker types on the binding affinity of ADCs to Fc γ R subtypes. Recognizing this gap, we selected four Fc γ Rs (I, IIa, IIb, and IIIa), which are critical for modulating ADCC, ADCP, and B-cell activation. Surface plasmon resonance (SPR) was employed to understand the binding affinity of the ISACs as compared to the corresponding naked mAb. In short, an anti-His-tag antibody was immobilized onto the carboxyl sensor using EDC/NHS chemistry. The appropriate His-tagged Fc γ R was loaded, followed by addition of a fixed concentration of the appropriate analyte. The concentration of analyte for each Fc γ R was based on a saturating concentration of the naked IgG1 mAb. In each case, the response (RU) for each analyte was normalized to the response of the naked mAb. (Supporting Table S1) Notably, and consistent with literature, deglycosylation of the naked antibody only partly ablated binding to the high-affinity Fc γ RI receptor (63% binding) while it completely ablated binding to the low affinity receptors Fc γ RIIa/b and Fc γ RIIIa (0% binding).³¹ Conjugation to the deglycosylated antibody further reduced binding to the high affinity receptor. Thus, all deglycosylated ISACs exhibited little or no measurable binding to any of the Fc γ R subtypes. However, the glycosylated ISACs, whether DAR8 or DAR2, exhibited comparable binding to the naked mAb across all Fc γ R subtypes.

In order to more fully understand the impact of conjugation on Fc γ R binding affinity, additional SPR studies were conducted using a range of analyte concentrations in order to establish the dissociation constant (K_D). (Supporting Figure S3 and Table 2) As anticipated, the Fc γ RI exhibited the strongest binding affinity to naked trastuzumab with a K_D of 12.7. Consistent with data shown in Table S1, conjugation to the endogenous cysteine residues had little or no impact on binding to Fc γ RI, as evidenced by the 9.6 nM and 5.7 nM affinities to the noncleavable and cleavable ISACs, respectively. Similarly, the K_D's for Fc γ RIIa, Fc γ RIIb and Fc γ RIIIa, while far weaker, were not significantly impacted by conjugation. The K_D's for the anti-HER2 mAb shown in Table 2 are comparable to reported IgG1 binding affinities for the shown Fc γ R subtypes.³² The K_D's for the deglycosylated ISACs were unable to be accurately determined because no measurable binding was observed at the maximal concentration tested (300 nM for Fc γ RI, 8000 nM for Fc γ RIIa/b, and 2000 nM for Fc γ RIIIa).

Fc γ R binding affinity is widely known to drive so-called effector function, largely in the form of antibody-dependent

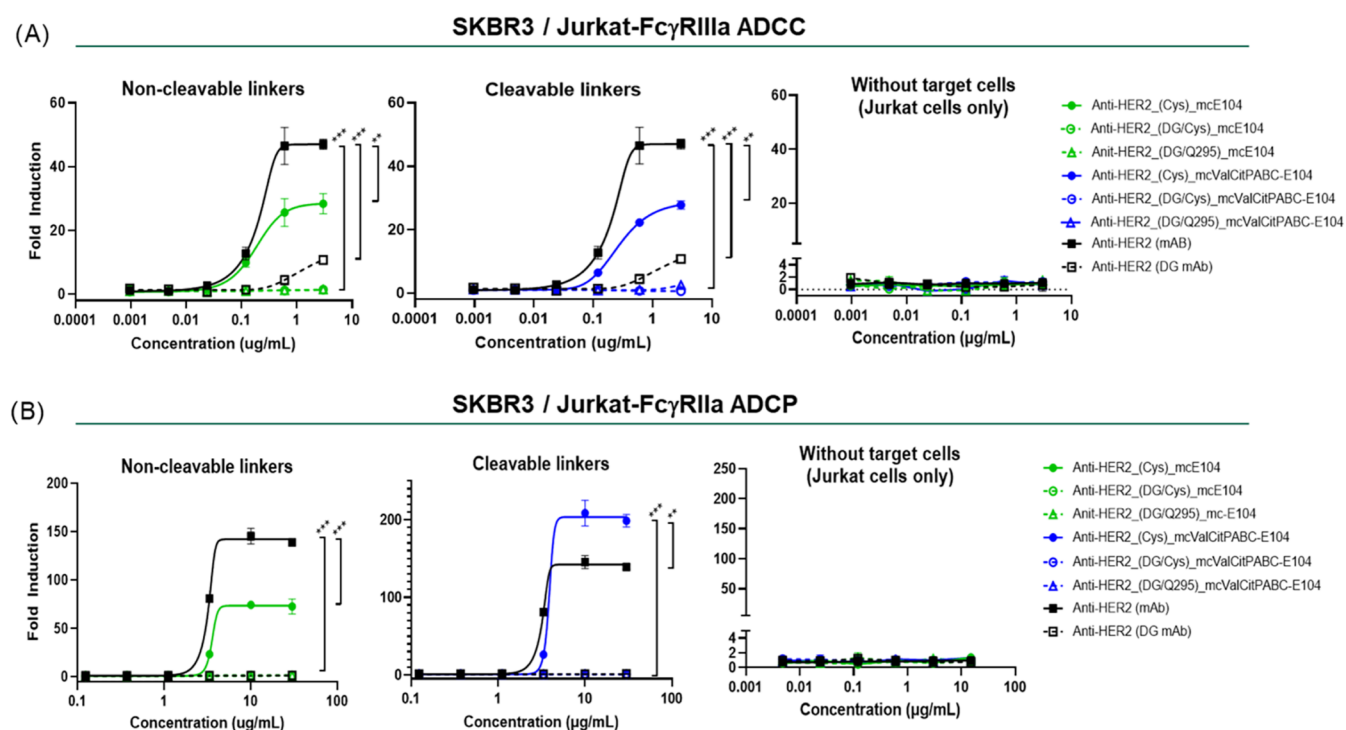


Figure 4. Assessment of ADCC and ADPC activity for ISACs and naked mAb. Glycosylated analytes are shown with a solid line and deglycosylated analytes are shown with a dotted line. (A) ISACs or mAbs were added to a coculture of Jurkat-Lucia NFAT-CD16a and SKBR3 cells ($\sim 2:1$) for 6 h. NFAT activation was assessed by induction of luciferase expression; (B) ISACs or mAbs were treated with a coculture of Jurkat-Lucia NFAT-CD32a and SKBR3 cells ($\sim 2:1$) for 6 h. NFAT activation was assessed by induction of luciferase expression.

cellular cytotoxicity (ADCC) or antibody-dependent cellular phagocytosis (ADCP). In order to further verify the SPR results and to understand the mechanism of action of our redesigned ISACs, it was important to directly assess effector function of our ISACs. With this in mind, we employed an ADCC coculture assay relying on Jurkat-Lucia NFAT-CD16a cells (Invivogen) cells that have been engineered to express a luciferase gene downstream of the Nuclear Factor of Activated T cells (NFAT) pathway. Interaction of Fc γ RIIIa (aka CD16a) with the antigen-bound antibody or ADC results in the dephosphorylation and nuclear translocation of NFAT thus triggering the expression of luciferase. In a similar manner, we employed an ADCP coculture assay relying on Jurkat-Lucia NFAT-CD32a cells (Invivogen) that overexpress Fc γ RIIa (aka CD32a). The reporter cells were incubated with the antigen-expressing SKBR3 cells for 6 h in the presence of the indicated ISAC, and the induction of the NFAT pathway was assessed by luciferase expression. The results are shown in Figure 4.

Most prominently, we found that the deglycosylated ISACs anti-HER2_(DG/Cys)_mcE104 and anti-HER2_(DG/Cys)_mcValCitPABC_E104 were completely devoid of ADCC and ADCP activity, even at concentrations as high as 15 μ g/mL (Figure 4). In contrast, the naked antibody exhibited both ADCC and ADCP activity with EC_{50} 's of ~ 0.3 μ g/mL and 3 μ g/mL, respectively. Removal of the glycosyl group from the naked antibody completely ablated ADCP activity, but some modest ADCC function was retained at concentrations higher than 1 μ g/mL. This indicates that both conjugation and removal of glycosylation is required for full ablation of ADCC function. Consistent with the SPR results discussed above, glycosylated ISACs exhibited comparable ADCC activity to the naked antibody, with both the cleavable anti-HER2_(Cys)_mcValCitPABC_E104 and the noncleavable anti-

HER2_(Cys)_mcE104 inducing NFAT activation at an EC_{50} of ~ 0.3 μ g/mL (Figure 4). As anticipated based on the weaker Fc γ RIIa binding affinity, both ISACs exhibited $\sim 10\times$ reduced EC_{50} 's in the ADCP assay (~ 3 μ g/mL) compared to the ADCC assay. Importantly, none of the ISACs or controls promoted NFAT activation in the absence of the SKBR3 target cells (Figure 4A,B, right-hand panels).

Efficacy of ISACs in an HCC1954 Xenograft Model.

Having established functional activity in a variety of coculture models (Figures 2, and 3) and having an adequate understanding of the effector function of the ISACs (Table 2, and Figure 4), we next set about comparing the efficacy of the various permutations of the E104 ISACs in a HER2+ xenograft model of breast cancer. As in the prior study (Figure 2), ~ 2 million HCC1954 cells were implanted in the right flank of 6-week-old female SCID/beige mice. Upon reaching a size of ~ 150 mm³, the animals were treated with 3 mg/kg of the appropriate ISAC (q4dx3, IP). As seen in Figure 5, nontreated animals exhibited rapid tumor growth and reached their humane end point in 10–20 days. The cleavable and noncleavable glycosylated ISACs were both efficacious, delaying tumor growth by 2 weeks or more (Figure 5A,B). Consistent with the prior results (Figure 2C), the deglycosylated cleavable anti-HER2_(DG/Cys)_mcValCitPABC-E104 promoted tumor regression or stasis in 5/7 animals (Figure 5B and Supporting Figure S6). Unexpectedly, based on the weaker activity in coculture models, the deglycosylated noncleavable anti-HER2_(DG/Cys)_mcE104 also exhibited significant activity, promoting tumor stasis or regression in 7/7 animals (Figure 5A and Supporting Figure S6). All of the Cys-conjugated ISACs promoted a statistically significant increase in survival as compared to vehicle control (Figure 5A,B).

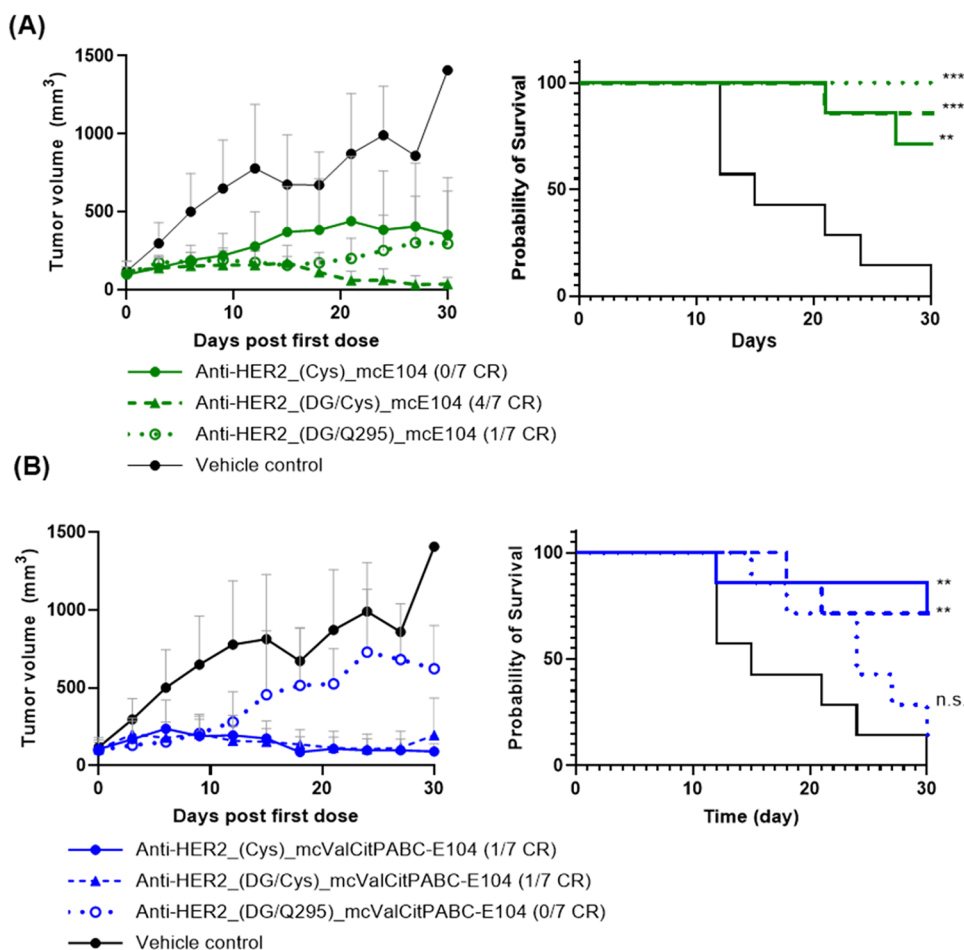


Figure 5. HER2-targeting immune stimulating ADCs were evaluated in a HCC1954 xenograft model using SCID beige mice. ($n = 7$ per group) Both noncleavable (green) and cleavable (blue) designs were evaluated. Dosing was performed at 3mpk by IP administration (q4dx3). (A) Tumor volume (left) and survival (right) of noncleavable ISACs in comparison to the vehicle control; (B) tumor volume (left) and survival (right) of cleavable ISACs in comparison to the vehicle control. Raw growth curves are shown in Supporting Figure S6. Statistical significance *vs* the vehicle control was established by Mantel-Cox test (* $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$).

The results for the site-specific Q295 conjugates (DAR2) were not entirely consistent with the DAR8 Cys-conjugated ISACs. The noncleavable **anti-HER2_(DG/Q295)_mcE104** (DAR2) exhibited a statistically significant increase in survival, nearly comparable to the DAR8 Cys-conjugated **anti-HER2_(DG/Cys)_mcE104**. (Figure 5A) Unexpectedly, however, the cleavable **anti-HER2_(DG/Q295)_mcValCitPABC-E104** conjugate exhibited very weak activity, only slightly increasing survival time compared to the vehicle control. (Figure 5B) The antitumor activity and increase in survival seen for **anti-HER2_(DG/Q295)_mcE104** is particularly interesting given the relatively low DAR, the lack of activity in coculture assays, and the lack of ADCC/ADCP activity. On the other hand, the lack of efficacy for **anti-HER2_(DG/Q295)_mcValCitPABC-E104** is consistent with the very poor activity in the coculture assays (Figure 3), and may be due to the significant decrease in DAR as compared to the matched cysteine conjugate.

Assessment of Pharmacokinetics, Immunogenicity, and Safety of E104-Based ISACs. While the efficacy shown in Figure 5 was encouraging, it was important to understand whether the differences in efficacy could be partially attributed to differences in PK exposure. In addition to exposure, we were also interested in understanding whether the ISACs may

induce immunogenicity *via* promotion of an Anti-Drug Antibody (ADA) response. ADA has been observed in various ISAC programs and is widely considered to be a risk factor for the development of immune-stimulating bioconjugates.¹⁷ With this in mind, we performed a multidose PK study using immune-competent C57BL/6 mice. The design of the study is shown in Figure 6A. As in our previous experiments, we included both glycosylated and deglycosylated naked antibodies as controls. The mice were administered a 3 mpk dose *via* intravenous injection, and blood samples were collected at 0.25, 6, 24, 48, 96, and 168 h. After a week of recovery, a second dose was administered, followed by blood collection at the same intervals and a terminal blood draw at the final point. In life observations such as body weight, animal behavior, and coat appearance, were used to assess adverse events associated with ISAC administration.

Both the glycosyl and deglycosylated naked antibody exhibited high total human antibody (tAb) exposure following the first dose, with an AUC_{0–168h} of 4000–6000 $\mu\text{g}\cdot\text{h}/\text{mL}$. While the deglycosylated antibody had a slightly lower exposure, the difference was within the error range of the assay. It is widely known in the antibody community that glycosylation has little impact on antibody clearance. However, all the DAR8 cysteine conjugates, both glycosylated and

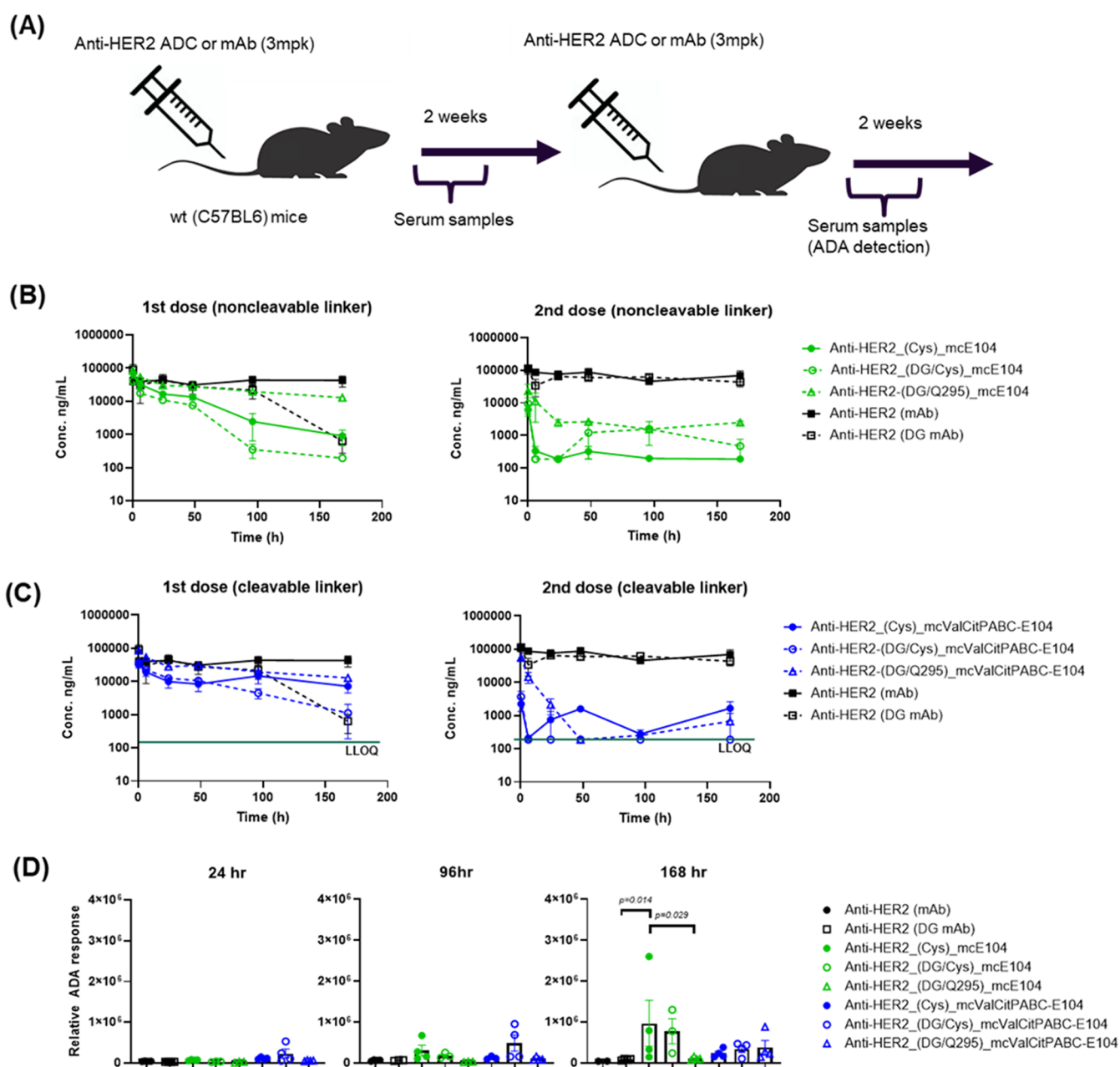


Figure 6. Multidose PK study was performed in order to understand the exposure and immunogenicity potential of the immune-stimulating ADCs. (A) Study design ($n = 4$ per group); (B) tAb exposure of noncleavable ISACs; (C) tAb exposure of cleavable ISACs; (D) relative antidrug-antibody (ADA) levels in serum at the shown time point following the 2nd dose of ISAC. Total antibody (tAb) was assessed by immunoassay. The limit of quantitation was ~ 200 ng/mL. ADA was measured by immunoassay using the ADC as the capture reagent and a goat-antimouse FLR647 as the detection reagent. Statistical analysis was performed using a Mann–Whitney U test.

deglycosylated, cleavable and noncleavable, exhibited significantly reduced exposure, typically 25–50% of the matched naked antibody. (Figure 6B,C and Table 3) For example, anti-HER2_(DG/Cys)_mcValCitPABC-E104 had an exposure of $1320 \mu\text{g}\cdot\text{h}/\text{mL}$ while the parent deglycosylated antibody had an exposure of $3870 \mu\text{g}\cdot\text{h}/\text{mL}$. It is widely known in the ADC field that high DAR conjugates often exhibit rapid clearance, likely due to uptake by Kupffer cells or scavenger receptors.^{33,34} Consistent with this hypothesis, we note that all the DAR8 cysteine conjugates exhibited a high HIC shift, generally 3–4 min longer than the corresponding naked antibody (Table 1). As anticipated, the Q295 conjugates (both DAR2) displayed much higher exposure, nearly equivalent to

the matched naked mAb (AUC $\sim 4000 \mu\text{g}\cdot\text{h}/\text{mL}$ for all 3), as might be anticipated based on their dramatically reduced HIC shift (1–2 min, Table 1). This suggests that the rapid clearance of the DAR8 conjugates after dose#1 results from the hydrophobicity of the ADC, a feature that can be modulated by various techniques such as site-specific conjugation and careful linker design.

Exposure of the naked antibody was higher after the second dose, likely due to incomplete clearance of the first dose at the time of administration. (See black lines in Figure 6B) However, exposure following the second dose of ISAC was dramatically lower than the first. (Figure 6B,C, Table 3) The cysteine conjugates, Anti-HER2_(Cys)_mc-E104, Anti-

Table 3. PK Exposure in C57BL/6 Mice Following Dose 1 and Dose 2 of Various ADCs and Control Antibodies^a

ADC	AUC ₀₋₁₆₈ Dose 1 (μg·h/mL)	AUC ₀₋₁₆₈ Dose 2 (μg·h/mL)
Anti-HER2 (mAb)	6750 ± 1420	11300 ± 2480
Anti-HER2 (DG mAb)	3870 ± 1240	9530 ± 1290
Anti-HER2_(Cys)_mcE104	1580 ± 298	56.0 ± 14
Anti-HER2_(DG/Cys)_mcE104	842 ± 106	189 ± 110
Anti-HER2_(DG/Q295)_mcE104	4640 ± 567	718 ± 166
Anti-HER2_(Cys)_mcValCitPABC-E104	1990 ± 605	158 ± 71.1
Anti-HER2_(DG/Cys)_mcValCitPABC-E104	1320 ± 172	42.0 ± 91.6
Anti-HER2_(DG/Q295)_mcValCitPABC-E104	4210 ± 574	425 ± 130

^aADCs and mAbs were dosed at 3mpk (iv).

HER2_(DG/Cys)_mcE104, Anti-HER2_(Cys)_mcValCitPABC-E104, and Anti-HER2_(DG/Cys)_mcValCitPABC-E104 all dropped below the limit of quantitation within just 6 h. (Note that the concentration seemed to inexplicably recover slightly for some ISACs by 48 h, but were still close to the limit of quantitation.) Interestingly, the Q295 conjugates, particularly the noncleavable anti-HER2_(DG/Q295)_mcE104, appeared to be cleared more slowly than the other ISACs—but still had a dramatic drop in AUC following the second dose (4640 vs 718 μg·h/mL).

The rapid drop in PK exposure following the second dose was unexpected and was strongly indicative of an ADA

response. Immune responses generated against therapeutic agents, including antibody-drug conjugates (ADCs), can adversely impact efficacy, safety, and pharmacokinetics. ISACs are designed to activate innate immune pathways, and in doing so can also heighten the risk of immunogenicity and ADA. ADA production may complicate therapeutic outcomes by neutralizing the ADC or exacerbating immune-related toxicities. ADA production is a known risk of immune-stimulating bioconjugates,¹⁷ but has not previously been reported in preclinical rodent models employing ISACs. In order to quantify ADA from our PK samples, we performed a sandwich immunoassay in which the ADC was immobilized using a biotinylated antihuman Fc, mouse plasma (from the PK samples) was incubated with the immobilized ADC, and detection of ADA was performed with a fluorescently labeled antimouse antibody. The relative ADA levels were measured at 24-, 96-, and 168 h postadministration of the second dose. (Figure 6D) These experiments clearly demonstrated the development of ADA following the administration of the ISACs but not the naked antibody comparators (Figure 6D).

In spite of the rapid clearance after the second dose, we observed no detectable ADA until 96 hpost-dose (Figure 6D). In order to explain this inconsistency, it is important to note that the sandwich immunoassay design will only detect free (unbound) ADA. Thus, it is unclear whether the increase in titer seen at 96 and 168 h is due increased ADA response or due to decreased circulating levels of ADC. Either way, it is

Table 4. In-Life Adverse Toxicity Assessment Following the Second Dose of ADC during the PK Study Described in Figure 6A^a

Test article	Animal	Body weight	Lethargy	Movement	Fur condition	Posture	Mortality	Total Score	Mean
Anti-HER2 (mAb)	M-1	0	0	0	1	0	0	1	1
	M-2	0	0	0	1	0	0	1	
	M-3	0	0	0	1	0	0	1	
	M-4	0	0	0	1	0	0	1	
Anti-HER2 (DG mAb)	M-1	0	0	0	1	0	0	1	1
	M-2	0	0	0	1	0	0	1	
	M-3	0	0	0	1	0	0	1	
	M-4	0	0	0	1	0	0	1	
Anti-HER2_(Cys)_mcE104	M-1	0	1	2	3	2	0	8	8
	M-2	0	1	2	3	2	0	8	
	M-3	0	1	2	3	2	0	8	
	M-4	0	1	2	3	2	0	8	
Anti-HER2_(DG/Cys)_mcE104	M-1	0	1	1	2	1	0	5	5
	M-2	0	1	1	2	1	0	5	
	M-3	0	1	1	2	1	0	5	
	M-4	0	1	1	2	1	0	5	
Anti-HER2_(DG/Q295)_mcE104	M-1	0	2	3	3	3	4	11	8.75
	M-2	0	1	2	3	2	0	8	
	M-3	0	1	2	3	2	0	8	
	M-4	0	1	2	3	2	0	8	
Anti-HER2_(Cys)_mcValCitPABC-E104	M-1	0	1	1	1	0	0	3	3
	M-2	0	1	1	1	0	0	3	
	M-3	0	1	1	1	0	0	3	
	M-4	0	1	1	1	0	0	3	
Anti-HER2_(DG/Cys)_mcValCitPABC-E104	M-1	0	0	0	1	0	0	1	1
	M-2	0	0	0	1	0	0	1	
	M-3	0	0	0	1	0	0	1	
	M-4	0	0	0	1	0	0	1	
Anti-HER2_(DG/Q295)_mcValCitPABC-E104	M-1	0	1	1	1	0	0	3	3
	M-2	0	1	1	1	0	0	3	
	M-3	0	1	1	1	0	0	3	
	M-4	0	1	1	1	0	0	3	

^aThe scoring system was developed based on the work of Flecknell and Morton. (REF) Higher scored indicate a more severe outcome. Scores were totaled for each animal and averaged for each treatment group, thus providing a rough measure of toxicity for each ADC. Note that no toxicity was observed following the first dose for any of the ISACs.

clear that the ADA response was most pronounced after exposure to the noncleavable **anti-HER2_(Cys)_mcE104** and **anti-HER2_(DG/Cys)_mcE104**. Interestingly, the analogous DAR2 Q295 conjugate (**anti-HER2_(DG/Q295)_mcE104**) promoted very little ADA response, consistent with the higher second dose PK exposure shown in Figure 6B. All three permutations of the cleavable (**mcValCitPABC-E104**) ISACs exhibited varying levels of ADA, but generally lower than the matched noncleavable ISAC. Notably, the focal point of our ISAC design, **anti-HER2_(DG/Cys)_mcValCitPABC-E104**, exhibited low, but measurable, levels of ADA.

A follow-on study was performed using the 168 h samples to understand if the observed ADA response was primarily against the antibody backbone or against the payload itself. Following a slight modification of the sandwich immunoassay mentioned above, the naked antibody (trastuzumab) was immobilized in place of the ISAC. The plasma sample was incubated with the immobilized antibody, and a fluorescently labeled antimouse antibody was employed as the detection reagent. The assay results revealed that ADA against the noncleavable **anti-HER2_(Cys)_mcE104** and **anti-HER2_(DG/Cys)_mcE104** particularly targets the antibody backbone. (Supporting Figure S4) This, of course, does not preclude the presence of ADA that targets the payload itself. The lower antibody-directed ADA signal for the cleavable ISACs was difficult to interpret in light of the reduced overall ADA signal for these analytes (Figure 6D).

During the course of the multidose PK study, we observed significant behavioral anomalies following the second dose of some of the ISACs. Building on the work of Morton,³⁵ we established a semiquantitative scoring system that would allow us to compare the observed toxicity of the various therapeutic agents. (Supporting Table S2) Specifically, we carefully noted changes in body weight, movement, fur condition, and posture for each animal, giving a score of 0 (good) to 4 (bad) for each criterion. (See Table S2 for details) As seen in Table 4, the noncleavable conjugates exhibited significantly worse scores than the matched cleavable conjugates. Specifically, most animals in these cohorts were found to have reduced movement around the cage, unkempt fur, and a crouched posture indicative of discomfort. Notably, one animal (M1) in the **anti-HER2_(DG/Q295)_mcE104** cohort had to be euthanized due to reaching humane behavioral end points. It is interesting to note that these behavioral anomalies were not present during the repeat-dose in the xenograft efficacy study that employed SCID/beige mice, suggesting that the toxicity is indeed immune-related. Encouragingly, however, all three cleavable ISACs exhibited very low scores, indicative of high tolerability. In particular, we were excited to note that the deglycosylated cleavable **anti-HER2_(DG/Cys)_mcValCitPABC-E104** conjugate displayed no signs of toxicity, and the behavior of the animals was indistinguishable from the naked mAb-treated cohort.

DISCUSSION AND CONCLUSIONS

The design of ADCs has become increasingly sophisticated over the past decade, now taking into consideration a plethora of factors, including linker polarity, site-of-conjugation, DAR, fucosylation, isoelectric point, rate of internalization, endosomal escape, and payload efflux. While challenging to balance all these factors, it is reasonably well understood how these various design factors influence the activity of ADCs that deliver cytotoxic payloads. However, as the field moves toward

noncytotoxic payloads,^{36,37} many of the “rules” that we have developed are being called into question. One of the areas where this is most clear is the field of immune-stimulating antibody conjugates (ISACs).

ISAC technology was born out of the success of checkpoint inhibitor therapeutics. The advent of immuno-oncology brought an awareness that the immune system can be “reprogrammed” in order to recognize cancer as foreign, thus taking advantage of a patient’s own adaptive immune system to engage with tumors. As such, ISAC technology is designed to selectively deliver immune-stimulating agents to tumor tissues in hopes of engaging our body’s innate and adaptive immune system in the fight against cancer. Due to this indirect mechanism of action, it is not clear that the traditional “rules” of ADC design will apply to the design of ISACs. Indeed, as an illustration of this point, nearly all published ISAC work directly claims that engagement with Fcγ receptors is necessary for activity.^{16–18,27,38,39} Moreover, the majority of published ISAC technology employs noncleavable linkers with the implicit assumption that they lack bystander activity and thus will be retained primarily in antigen-presenting myeloid cells after ADCP-mediated uptake. Our study herein is designed to test these assumptions in order to provide critical data for next-generation ISAC programs.

We focused our studies on a specific TLR7 agonist known as “E104” that our team reported in 2022. TLR7 agonists are widely known as vaccine adjuvants, specifically activating innate immune pathways present in B-cells, dendritic cells, and myeloid cells. Unlike many toll-like receptors, the expression profile of TLR7 is rather narrow, only being expressed in narrow subsets of immune cells, thus reducing the potential for systemic toxicity (*i.e.*, cytokine release syndrome). Indeed, several systemically delivered TLR7 agonists have entered clinical development⁴⁰—although none have reached approval status. While many tumor-targeting TLR7 agonist designs have been reported in the literature, the work by Ackerman in 2020 promoted a paradigm shift by using ADC technology to deliver TLR7 agonists specifically to tumor tissue. This work ultimately led to two clinical-stage ISACs (BDC-1001 from Bolt Biotherapeutics and NJH-395 from Novartis) that employed a noncleavable linker and a typical FcγR-intact antibody.^{16,17} Building on this work, we prepared a set of E104-based tumor-targeting and isotype control ISACs that systematically varied glycosylation status, linker type, and site of conjugation. (Table 1) These conjugates were advanced through a series of functional tests to understand their potency, mechanism of action, and therapeutic potential.

The initial focus of our efforts centered on an E104-based ISAC employing a cleavable linker and a deglycosylated (DG) antibody backbone, **anti-HER2_(DG/Cys)_mcValCitPABC-E104**. We hypothesized that the cleavable linker and highly permeable payload would allow for ablation of FcγR binding affinity without compromising the antitumor efficacy. While this initial design was reported in our 2022 publication,²² at that time we did not compare this design head-to-head with the industry standard ISAC design (“NJH395-like”) that used a glycosylated Fc domain (FcγR-intact) and a noncleavable linker. Consistent with our prior reports, we found that our redesigned ISAC exhibited potent immunostimulatory activity in coculture assays with mouse and human immune cells and excellent antitumor efficacy in a HER2+ xenograft model. (Figure 2) Both the *in vitro* and *in vivo* activity were comparable or better than the industry-standard ISAC

construct. Importantly, we found that the traditional ISAC design induced antigen independent activity, promoting IL-6 and TNF α release in PBMCs (Supporting Figure S2) and modest antitumor activity of the isotype-control ISAC (Figure 2C). In contrast, the redesigned Fc γ R-ablated/cleavable linker ISAC did not promote cytokine release in PBMC monoculture and the isotype-control ISAC was devoid of antitumor activity.

Having established the validity of our new ISAC design, we next studied a systematically prepared set of ISACs that varied in terms of their glycosylation state (glycosylated/deglycosylated), linker design (cleavable/noncleavable), and drug loading (DAR8/DAR2). Most notably, we observed that ISACs employing a noncleavable linker (*i.e.*, Anti-HER2_(DG/Cys)_mcE104) exhibited significantly lower potency than their cleavable counterparts in coculture assays using HER2+ cancer cells and mouse macrophages. (Figure 3A) This same phenomenon was not observed in mouse splenocyte cocultures. (Figure 3B) The efficacy of the noncleavable ISACs in a HER2+ xenograft study (Figure 5) suggests that the splenocyte coculture assay is more predictive of the *in vivo* efficacy than the macrophage coculture assay.

Similarly, the lack of efficacy of the DAR2 anti-HER2_(DG/Q295)_mcValCitPABC-E104 in mouse splenocytes was predictive of its lack of efficacy in the xenograft model. (Figures 3B vs 5B) While the lower DAR is a possible explanation for this phenomenon, this seems unlikely due to the dramatic loss in potency (~ 30 – $100\times$) as compared to the relatively modest loss in DAR ($\sim 4\times$). Thus, a more plausible explanation lies with the rate of linker cleavage. In our prior work on Q295 site-specific conjugation, we demonstrated that the rate of Q295 linker cleavage was dramatically reduced as compared to cleavage at endogenous cysteine residues.²⁵ Given the high permeability of E104, we hypothesize that it simply diffuses into bulk solvent (coculture assays) or diffuses out of the tumor microenvironment (xenograft model) before it can suitably activate the tumor-associated myeloid cells. In contrast, higher DAR conjugates attached to cysteine sites (anti-HER2_(Cys)_mcValCitPABC-E104 and anti-HER2_(DG/Cys)_mcValCitPABC-E104) result in a burst of payload release upon internalization that allows for a high localized concentration of payload that drives myeloid cell activation.

Perhaps most importantly for the present study, we set out to conclusively demonstrate whether or not Fc γ R-mediated effector function is required for ISAC efficacy. As mentioned above, several currently published studies have claimed that ISACs rely on Fc γ R-mediated uptake of opsonized target cells to drive payload release and activation of myeloid cells and APCs.^{16,17,27} Herein, we have provided clear evidence against this hypothesis. As shown in Table 2 and Figure 4, none of the deglycosylated ISACs exhibit any measurable ADCC or ADCP activity at the highest tested concentrations, and yet several of the deglycosylated ISACs exhibit excellent potency (0.3–3 μ g/mL) in immune-cell coculture and mouse xenograft models. Most notably, anti-HER2_(DG/Cys)_mcValCitPABC-E104 displayed robust efficacy in two xenograft models (Figures 2C and 5), potentially induced activation of the NF κ B and IRF pathways in cocultured myeloid cells (Figure 3A,B), and promoted the release of IFN α from cocultured human PBMCs. Taken together, this strongly suggests the mechanism shown in Figure 1, wherein the payload is released in antigen-expressing tumor cells (TLR7 negative) and activates adjacent TLR7-positive myeloid cells in a bystander-like mechanism.

Apparently, one advantage of this “bystander-like” mechanism is reduced toxicity. The study outlined in Table 4 shows that cleavable ISACs exhibit very little toxicity in a repeat-dose study in immunocompetent mice. In contrast, the noncleavable ISACs induced a number of dose-limiting in-life findings suggestive of significant toxicity. While the study design did not permit a mechanistic understanding of these findings, it is apparent that these noncleavable ISACs promote adverse events in immune-competent (C57BL/6) mice that were not observed in the efficacy study employing immune-compromised SCID beige mice (Figure 5). The reason for these findings remains unclear at the moment, but may relate to the uptake and retention of a highly potent metabolite, Cys-mcE104. (Supporting Figure S5) In a preliminary study of this metabolite, we found that it activates TLR7-expressing Ramos cells at comparable or higher levels than E104 itself. Analogous to the well-studied mcMMAF catabolite, the Cys-mcE104 catabolite is expected to be highly polar with limited permeability. Plausibly, the concentration of this polar metabolite builds up over time in myeloid cells, resulting in overactivation of various immune-stimulatory pathways that drive the observed efficacy (Figure 5), but also promoting toxicity and ADA response. Further investigations will be necessary to untangle the mechanism of toxicity and efficacy for the Fc-ablated noncleavable ISACs.

Likewise, it is interesting to speculate about the unexpected efficacy of the Fc-ablated noncleavable ISACs (Anti-HER2_(DG/Cys)_mcE104 and Anti-HER2_(DG/Q295)_mcE104). Both of these conjugates promoted the release of IL-6 in a coculture between HCC1954 cells and mouse splenocytes, suggesting a non-Fc γ R-mediated process for splenocyte activation. (Figure 3B) In line with the aforementioned *in vitro/in vivo* correlation of this assay, we note a robust antitumor response for each of these bioconjugates. As noted in Supporting Figure S5, the cysteine adduct of mcE104 has a Clog *P* of 2.97 and a log *D*_{7.4} of 0.85, suggestive of modest permeability. Thus, it is conceivable that the efficacy of the deglycosylated mcE104 conjugates is, in fact, driven by bystander activity. Additional studies will be reported on the mechanism of these noncleavable Fc-ablated ISACs will be reported in due course.

It is interesting to note that most of the E104-based ISACs exhibit efficacy in spite of a relatively modest (or even poor) pharmacokinetic profile. The exposure (AUC) after the first dose of DAR8 cysteine-based ISAC was only 25–50% of the matched naked mAb. (Table 3) The deglycosylated ISACs display marginally lower exposure than their matched glycosylated counterpart. However, the primary driver of poor exposure is almost certainly the high drug-antibody ratio (DAR), a well-known factor in ADC pharmacokinetics. In this respect, it is important to point out the increased exposure of the matched Q295 (DAR2) conjugates, which nearly match the exposure of the corresponding naked antibody. This increased exposure may be responsible for the unexpected efficacy of the Anti-HER2_(DG/Q295)_mcE104, in spite of its modest potency in coculture assays (Figure 5A).

Regardless of the differences in PK exposure after the first dose, it was clear that all ISACs were rapidly cleared after the second dose, strongly suggestive of an ADA response. This was confirmed using a sandwich ELISA assay in which the ADC itself was used as the capture reagent and a goat-antimouse antibody was used as the detection reagent. (Figure 6D) As anticipated, the highest ADA signals were evident ~ 1 week

after the second dose. Importantly, the naked antibody did not induce an ADA response, in spite of the mismatch in species (human antibody dosed in mice). This indicates that the TLR7 agonist is acting as an adjuvant, inducing an immune response against the conjugate itself. Future studies will repeat this study in tumor-bearing mice in order to assess the production of antitumor antibodies. It should be noted that the response appears to be primarily directed against the antibody backbone, as very similar results were observed when using trastuzumab as the capture reagent in place of the ISAC. (results not shown) Thus, it is unclear whether the ADA response is translationally relevant. For example, perhaps the ADA response would be dramatically reduced if the TLR7 agonist were conjugated to a mouse IgG1 instead of a human IgG1. Either way, to our knowledge, this is the first published study of an immune-stimulating ADC to conclusively demonstrate an ADA response in mice.

In spite of these challenges associated with ADA, our study clearly shows that Fc-ablation does not impede the efficacy of E104-based ISACs. Moreover, our results suggest that combining a cleavable linker with Fc γ R ablation may result in improved tolerability. This is an important finding that opens up opportunities for new immune-stimulating ADC designs. Additionally, our study suggests an *in vitro/in vivo* correlation between a mouse splenocyte coculture assay and a xenograft efficacy model. Finally, this study identifies an immunogenicity (ADA) risk of ISACs in mouse models that warrants futures study. The focal ISAC of this study, **anti-HER2 (DG/Cys)_mcValCitPABC-E104** exhibited robust efficacy in an HCC1954 (HER2+) xenograft model, good tolerability in both immune competent and immune compromised mice, and strong induction of IFN α in hPBMC cocultures. Future studies of this technology will attempt to apply this linker-payload to ISAC programs beyond anti-HER2 and will focus on developing a deeper understanding of the efficacy and safety in immune-competent (wild-type) mice.

EXPERIMENTAL DETAILS

General. Antibodies (trastuzumab and palivizumab) were custom prepared by Syd laboratories (Hopkinton, MA) and were analyzed by LC-MS and SEC prior to use. The payload (E1104) and the cleavable linker-payload **mcValCitPABC_E104** was prepared as previously described.²² Anti-TNF α anti-IFN γ , and anti-IL-6 DuoSet ELISA kits were purchased from R&D systems. Anti-IFN α ELISA kit was purchased from PBL biosciences. QuantiBlue and QuantiLuc detection kits were purchased from InvivoGen. SKBR3 HCC1954 cells were purchased from ATCC, Ramos Blue cells and RAW dual cells were purchased from InvivoGen, and human PBMCs were purchased from iXCells. Mouse splenocytes from C57BL/6 mice were purchased from Fisher. Cell lines were maintained as recommended by the supplier and passaged every 3–4 days. Cells used for all assays were below passage 25. Cell culture supplies, including culture media, FBS, and penicillin/streptomycin were purchased from VWR. SCID Beige mice (CB17.Cg-Prkdc^{scid}Lysf^{bg-j}/CrJ) and C57BL/6 mice were purchased from Charles River Laboratories. Implantation matrix Matrigel was purchased from VWR. All synthetic compounds were confirmed to be >95% pure by HPLC.

Statistical Methods and Rigor. The number of biological and technical replicates for each experiment is indicated in the corresponding figure legends. Generally, all cell culture experiments were performed in biological triplicate and results are plotted as the mean and standard deviation. *In vitro* experiments were not randomized, and investigators were not blinded to sample allocation or outcome assessment during these studies. For all *in vivo*

experiments, the exact number of mice per treatment group is provided in the figure legends. Animals were randomized into groups based on tumor size prior to treatment initiation. Investigators conducting *in vivo* studies were blinded to group allocation during both treatment and data collection. All statistical analyses were performed using Microsoft Excel or GraphPad Prism. Comparisons between two groups were conducted using either paired or unpaired two-tailed Student's *t* tests, as appropriate. For comparisons involving more than two groups, one-way or two-way analysis of variance (ANOVA) was applied, followed by post hoc multiple comparisons using Tukey's correction. Statistical analysis of immunogenicity samples was performed using a Mann–Whitney U test due to the large intraindividual differences in immune responses to the test articles. A *p*-value less than 0.05 was considered statistically significant. LCMS was performed as previously described.²²

Preparation of *N*-(4-((4-Amino-2-butyl-1*H*-imidazo[4,5-*c*]quinolin-1-yl)methyl)phenyl)-6-(2,5-dioxo-2,5-dihydro-1*H*-pyrrol-1-yl)hexanamide [mcE104]. A solution of 1-(4-amino-benzyl)-2-butyl-1*H*-imidazo[4,5-*c*]quinolin-4-amine (**E104**) (10.0 mg, 29 μ mol), 6-(2,5-dioxo-2,5-dihydro-1*H*-pyrrol-1-yl)hexanoic acid (7.3 mg, 34.74 μ mol), HATU (14 mg, 38 μ mol), HOBt (6.2 mg, 40 μ mol) and 2,6-dimethylpyridine (9.3 mg, 10.1 μ L, 87 μ mol) was stirred at rt for 1.5 h. The crude mixture was purified by preparative HPLC to obtain the desired product (11.2 mg, 71.8%) as a faint yellow solid. LC-MS: *m/z* 538.6 [*M* + 1]⁺; Retention time = 3.47 min. ¹H NMR (400 MHz, DMSO-*d*₆) δ _H/ppm 9.86 (s, 1H), 7.96 (d, *J* = 8.1 Hz, 1H), 7.79 (d, *J* = 8.2 Hz, 1H), 7.65–7.60 (m, 1H), 7.52 (dd, *J* = 9.0, 2.9 Hz, 3H), 7.40–7.35 (m, 1H), 7.01 (d, *J* = 4.0 Hz, 2H), 7.00–6.98 (m, 2H), 6.97 (s, 2H), 6.96 (d, *J* = 2.3 Hz, 1H), 5.89 (s, 1H), 3.37 (dd, *J* = 8.8, 5.3 Hz, 3H), 2.98 (d, *J* = 7.5 Hz, 1H), 2.95 (d, *J* = 3.2 Hz, 1H), 2.24 (dd, *J* = 12.8, 5.4 Hz, 2H), 1.72 (ddd, *J* = 12.2, 7.4, 3.6 Hz, 2H), 1.58–1.46 (m, 3H), 1.38 (dt, *J* = 14.5, 7.5 Hz, 2H), 1.27–1.17 (m, 2H), 0.85 (t, *J* = 3.7 Hz, 2H).

Interchain Disulfide Conjugation. The appropriate antibody was diluted to 2 mg/mL in PBS containing 5 mM EDTA and subsequently treated with TCEP (7–8 mol equiv). After incubation for 2 h at 37 °C, 15 equiv of the appropriate linker-payload was added as a 10 mM stock solutions in DMA along with sufficient DMA to guarantee solubility (10% total organic for mcE104 and 15% total organic for mcValCitPABC-E104). After 2h incubation at room temperature, the reaction was buffer exchanged into 2 mL of DPBS using a PD-10 Sephadex column according to the manufacturer's protocol. The drug-antibody ratio (DAR) was determined by LCMS and the amount of aggregation was established by SEC. Hydrophobicity was established by HIC. (see [Supporting Details](#) for analytical methods) The final ADC was filter sterilized through a 0.22 μ m filter prior to storage. Endotoxin levels were assessed using the ToxinSensor Chromogenic LAL Endotoxin Assay Kit (GenScript) and was verified to be below 1 EU/mg.

Conjugation to the Q295 Site. Step 1: Conjugation to Cystamine. The appropriate antibody (10 mg) was treated with 2.5 μ g of PNGase F (Bulldog Bio) and diluted to a final concentration of 10 mg/mL in PBS. The antibody was incubated at 37 °C overnight. LCMS analysis indicated that the deglycosylation was complete. The deglycosylated antibody was diluted with 2.4 mL of 50 mM phosphate buffer (pH 6.0) and treated with 200 μ L (100 equiv) of 30 mM aqueous cystamine solution followed by 700 mg of transglutaminase powder (Ajinomoto, Activa). The reaction was vortexed extensively until the transglutaminase powder was completely dissolved. The reaction was allowed to stand at 37 °C for 48 h. The reaction was loaded onto 10 prewashed protein A SpinTrap columns from Thermo Fisher (~1 mg/column). After washing extensively with PBS to remove the excess transglutaminase and cystamine, the modified antibody was eluted with 400 μ L of glycine buffer (pH 2.7) and immediately neutralized with 30 μ L of 1 M Tris buffer (pH 9.0). The combined eluents were buffer exchanged into PBS by ultrafiltration (10 kDa cutoff filter). Loading was evaluated by LCMS.

Step 2: Conjugation of the Linker-Payload to the Thiolated Q295 Site. The cystamine conjugated mAb from step 1 was reduced with 4–8 mol equiv of triphenyl phosphine meta-sulfonic acid

(TPPMS) for 2–4 h. After the reduction, 10–12 equiv of mc-E104 or mcValCitPABC-E104 was added (as a DMA stock solution) for the final 4-h conjugation step. The percentage of DMA in the reaction system was controlled to 10%–15% to maintain the solubility of the linker payloads. The ADCs were buffer-exchanged into DPBS using 10 kDa spin filter columns (Sartorius/VS0102) or PD-10 Sephadex columns according to the manufacturer's protocol. The ADC solution was then filter sterilized and characterized by A280, SEC, HIC, and LCMS. (see [Supporting Details](#) for analytical methods)

SKBR3/RAW Dual and HCC1954/RAW-Dual Coculture Assay. HCC1954 cells, SKBR3 cells, and RAW-dual cells were cultured according to the manufacturer's instructions in high-glucose DMEM supplemented with 10% FBS. SKBR3 cells or HCC1954 cells were premixed 1:1 with RAW-dual cells giving a final seeding density of 0.2 million cells per milliliter each. 90 μ L of the cell mixture was seeded into a 96-well plate and treated with 10 μ L of the appropriate test article (ISAC, mAb, or DPBS). ISACs and naked mAbs were diluted to 10 $\times$ the desired final concentration. The cells were cultured for 48 h at which point the supernatant was separated *via* centrifugation, and the activation of the NF κ B pathway and IRF pathway was determined using the QuantiBlue and QuantiLuc assay, respectively. For the QuantiBlue assay, 20 μ L of the supernatant were added to 180 μ L of freshly made QuantiBlue solution prepared according to manufacturer's protocol and incubated for 4 h. Absorbance at 450 nm measured using a VarioSkan microplate reader. For the QuantiLuc assay, 20 μ L of the supernatant were added to 50 μ L of freshly thawed QuantiLuc solution prepared according to manufacturer's protocol. All-wavelength luminescence was immediately measured using a VarioSkan microplate reader. In each case, the observed signal was normalized to that of a nontreated control.

HCC1954/hPBMc Coculture Assay. HCC1954 cells were maintained in RPMI 1640 media supplemented with 10% FBS. Cells were harvested, rinsed, and resuspended into a 0.2 million per mL seeding suspension. 90 μ L of the cell mixture was seeded into a 96-well plate and incubated for 2 h to allow stable attachment. On assay day, frozen human PBMcs were thawed, rinsed, and resuspended into a 2 million per mL suspension. Culture media in the HCC1954 plate was discarded and 100 μ L of PBMc suspension was added to each well. 100 U/mL penicillin and 100 μ g/mL streptomycin were applied to prevent contamination. ISACs were diluted to provide stock solutions 10 $\times$ higher than the test concentrations and 10 μ L of each was added to the PBMc suspension. The cells were cultured for 24 h at which point the supernatant was separated *via* centrifugation. IFN α levels in the supernatant was measured using a pan-IFN α ELISA kit (PBL Assay Sciences). The data was analyzed and plotted using GraphPad Prism 7 software.

SKBR3/Mouse Splenocyte and HCC1954/Mouse Splenocyte Coculture Assay. SKBR3 and HCC1954 cells were maintained in RPMI 1640 media supplemented with 10% FBS. On the assay day, cells were harvested, rinsed, and resuspended to a suspension of 0.1×10^6 cells/mL per mL. 100 μ L of the SKBR3 or HCC1954 seeding suspension was added to a 96-well plate and incubated overnight to allow adherence. Frozen mouse splenocytes were thawed, rinsed, and resuspended into a 1.5 million per mL seeding suspension. On assay day, the culture media in the SKBR3/HCC1954 plate was discarded and 100 μ L of mouse splenocyte suspension was added. Ten μ L of the appropriate test article was added, reaching a final concentration of 50, 10, 2, 0.4, 0.08, and 0 μ g/mL. The cells were cocultured for 24 h. After the incubation, supernatant samples were isolated *via* centrifugation. IL-6 in the supernatant was measured by a DuoSet mouse IL-6 ELISA kit (R&D systems). Data was analyzed and plotted using GraphPad Prism 7 software.

Xenograft Efficacy Study. (Figure 2C) Human breast cancer cells HCC1954 were cultured in RPMI 1640 supplied with 10% FBS and maintained under the manufacturer's recommended densities. 100 U/mL penicillin–streptomycin was applied to prevent microbe contamination. Immediately before the tumor implantation, the cells were trypsinized, rinsed, and resuspended into a suspension of 40 million cells per mL. The suspension was mixed 1:1 with Matrigel (Corning) to form the final implantation mixture, which was kept on

ice for no longer than 2 h. Approximately 2 million cells (100 μ L of the mixture) were implanted subcutaneously to the right flank of 6–8 week old female SCID beige mice (Charles River Laboratories). Tumor volume was recorded twice a week using a caliper and estimated using the following formula: length $\times$ width²/2. ADC treatment was initiated once the tumor volumes reached 50–300 mm³. Test articles were administered *via* intraperitoneal injection 3 times in total with 5-day intervals. Tumor volumes and body weights were measured every 2–3 days. Mice whose tumor exceeded 1000 mm³, suffered from ulceration, or displayed any signs of stress during the study were euthanized based on IACUC-approved animal protocols. All ADCs were administered as PBS formulations.

Xenograft Efficacy Study. (Figure 5) The study was conducted as above except dosing was performed q4dx3 (3mpk) and tumor implantation was performed using basement membrane extract #3 (R&D systems) in place of Matrigel.

Fc γ R Binding Assay. The SPR-based binding assay was carried out using a Nicoya Open SPR at 25 $^{\circ}$ C with PBS as running buffer. A high-capacity carboxyl sensor was prepared by injection of a 50 mM solution of 1:1 NHS and EDC at 20 μ L/min. The AntiHis tag antibody (Genetex/GTX40628) was diluted to 20 μ g/mL in 200 μ L of pH 5.0 acetate buffer and was immobilized on Channel 1 and 2 using a 20 μ L/min flow rate. Any remaining carboxyl groups were blocked with 1 mM ethanolamine. The appropriate histidine-tagged recombinant Fc γ R was diluted in PBS to 6 μ g/mL and loaded onto the channel 2 sensor at 10 μ L/min. The test samples were diluted in running buffer to a concentration correlating with the previously reported max binding concentration of a standard mAb to the individual Fc γ R.⁴¹ Analytes were injected at 20 μ L/min and at least 5 min were allowed for disassociation. A 10 mM glycine-HCl buffer (pH 1.5) was injected twice at 150 μ L/min to remove Fc γ R and analyte prior to the next analysis. Nicoya software was used to collect the raw binding data and analysis was performed using Tracedrawer. Fc γ RI (Sino Biologicals/10256-H08S-100, max concentration = 300 nM); Fc γ RIIa (Sino Biologicals/10374-H08H1-1, max concentration = 8000 nM); Fc γ RIIb (Sino Biologicals/10259-H08H-100, max concentration = 8000 nM); Fc γ RIIIa (Invitrogen/RP87975, max concentration = 2000 nM). For maximum RU measurements, all test articles were evaluated at the max concentration shown above. For K_D determination, all test articles were tested at the max concentration shown above and diluted in 1:3 in PBS for a total of 6 concentrations.

ADCC Reporter Assay. Jurkat-Lucia NFAT-CD16 cells (InvivoGen/part ID: jknl-nfat-cd16) were cultured according to the manufacturer's instructions and passaged every 2–3 days in IMDM growth media. Prior to the assay, the cells were centrifuged at 300g for 5 min and resuspended in IMDM test media (2 mM L-glutamine, 25 mM HEPES, 10% heat-inactivated FBS, 1% Pen-Strep without antibiotics) at a concentration of $\sim 5 \times 10^5$ cells/mL. The cells were incubated at 37 $^{\circ}$ C in 5% CO₂ for 48 h. SKBR3 cells (ATCC Part ID: HTB-30) were passaged every 3–4 days in McCoy's 5A media. Approximately 12h before the assay, SKBR3 cells were trypsinized then centrifuged at 200g for 5 min and resuspended in McCoy's 5a media (1.5 mM L-glutamine, 2200 mg/L sodium bicarbonate, 10% heat-inactivated FBS, 1% Pen-Strep) at a concentration of 1.1×10^6 cells/mL. Ninety microliters of the SKBR3 cells (~ 100 k cells) were added to a treated 96-well clear flat-bottom plate. The plate was incubated at 37 $^{\circ}$ C in 5% CO₂ overnight. Serial dilutions of antibodies and ADCs were prepared (5 $\times$ dilution series) ranging from 30–0.0096 μ g/mL. Before adding the test article, McCoy's 5a media was removed from each well and replaced with 90 μ L of IMDM test media. Twenty microliters of the antibody or ADC were added to each well. The plate was incubated at 37 $^{\circ}$ C in 5% CO₂ for 1 h. Jurkat-Lucia NFAT-CD16 cells were resuspended in IMDM test media at a concentration of $\sim 2.2 \times 10^6$ cells/mL, and 90 μ L of these cells were added to each well (~ 200 k cells/well). The plate was incubated at 37 $^{\circ}$ C in 5% CO₂ for 6 h. After incubation, 20 μ L of the supernatant from each well was transferred to a white opaque 96-well plate, and 50 μ L of QUANTI-Luc 4 Lucia/Gaussia working solution was added to each well. Luminescence was measured immediately using a Varioskan LUX 50.

ADCP Reporter Assay. The Jurkat-Lucia NFAT-CD32 (InvivoGen/part ID: jk1-nfat-cd32) was performed as above except that the test articles were prepared using a 5-fold serial dilution ranging from 150–0.048 $\mu\text{g/mL}$.

Multidose Mouse Pharmacokinetic Study. C57BL/6 mice from Jackson Laboratory were divided into nine groups of four mice each. Test articles were dosed by tail vein IV administration at 3 mg/kg using a PBS formulation. Blood samples were collected into heparinized tubes *via* cheek puncture from the saphenous vein at 0.25, 6, 24, 48, 96, and 168 h from each mouse. Plasma was obtained by centrifugation and stored at $-80\text{ }^{\circ}\text{C}$ until analysis. After the 168h blood draw, the animals were allowed to recover for 1 week, and then a second dose of the same test article was administered (IV, 3 mg/kg). Plasma samples were again obtained at the specified time intervals. The animals were sacrificed at 168 h after the second dose by asphyxiation, and a terminal blood draw was taken. Animals were carefully monitored for signs of adverse events using the scoring system outlined in Table S2. Total human IgG was assessed as described below. The area-under-the-curve (AUC) for 0–168 h values were calculated using noncompartmental analysis conducted with Phoenix WinNonlin software (version 8.40, Princeton, NJ).

Quantitation of total human IgG. Total antibody quantitation was performed using a Corria PIXI automated ELISA system with the Human IgG detection kit (KPX1-PCEA101–001). Mouse serum samples from the PK study were thawed at room temperature. 2 μL was removed and diluted 10 $\times$ in mouse serum (Innovative Research Inc./IMSNSASER100 ML). The standard curve was prepared by diluting each test article using mouse serum as diluent (5 $\times$ dilutions) with final concentrations ranging from 200,000 to 2.56 ng/mL. Each mouse serum test sample and the appropriate serial dilution were diluted 10-fold into PIXI sample diluent (Part ID: SD-010) and used within 1 h. The test article was loaded onto a cartridge precoated with an anti-Human Fab following the manufacturer's instructions. After the test article was loaded, an FLR-647 antihuman mAb (Part ID: PRI-2IGG) was diluted to 1:25 in wash buffer (Part ID: WB-008) and used as the detection antibody. The setup and analysis were performed according to the manufacturer's instructions. Analysis was performed using the PIXI software and graphed using Graphpad Prism.

Detection of Mouse Antidrug Antibodies (ADA). Frozen serum samples from 24, 96, and 168 h after the second dose were thawed at room temperature and diluted 10 $\times$ into Correlia sample diluent. The dosed ISAC or mAb (the capture reagent) was diluted to 100 $\mu\text{g/mL}$ in sample diluent and immobilized on the cartridge that had been precoated with antihuman Fab. An antimouse antibody labeled with FLR-647 (Jackson ImmunoResearch, Cat# 115–605–205) was diluted to 30 $\mu\text{g/mL}$ in Corella Wash buffer and used as the detection antibody.

ADA Targeted to Naked mAb. The method described above was modified by replacing the ISAC capture reagent with naked anti-HER2 (trastuzumab) to detect ADA specifically targeted to the antibody portion of the ISAC (as opposed to the payload/conjugation).

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jmedchem.5c01908>.

Experimental details, LCMS methods, HER2-targeted ISACs exhibit no direct cytotoxicity against HER2+ SKBR3 and HCC1954 cells, SPR assessment of glycosylated ISACs as compared to the naked mAb, relative binding of various ISACs to the shown Fc γ R subtypes, and raw characterization data for the ISACs/ADCs (PDF)

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Notes

Ethical statement All animal studies followed ACS Ethical Guidelines and were approved and overseen by the Binghamton University Institutional Animal Care and Use Committee (IACUC). Studies were conducted according to IACUC approved protocol 863–21 and 890–23. This research did not involve human participants.

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■ ABBREVIATIONS USED

ADC, antibody-drug-conjugate; ADCC, antibody-dependent cellular cytotoxicity; ADCP, antibody-dependent cellular phagocytosis; ADA, antidrug-antibodies; APCs, antigen-presenting cells; Cys, cysteine; DPBS, Dulbecco's phosphate

buffered saline; DAMPs, danger-associated molecular patterns; DAR, drug-to-antibody-ratio; DG, deglycosylated; HIC, hydrophobic interaction chromatography; ICI, immune checkpoint inhibitors; IRF, interferon response factor; ISACs, Immune Stimulating Antibody Conjugates; LBA, ligand-binding assay; MC, maleimide caproyl; mcValCitPABC, maleimide caproyl valine citrulline para-amino benzyl carbamate; mAb, monoclonal antibody; NF κ B, nuclear Factor kappa-light-chain-enhancer of activated B cells; NFAT, nuclear factor of activated T-cells; PRR, pattern-recognition receptors; PBMC, peripheral blood mononucleocyte; tAb, total antibody; TAMs, tumor-associated macrophages; TCEP, (tris(2-carboxyethyl)-phosphine); TLR7, toll-Like Receptor 7; TPPMS, triphenylphosphine-*meta*-sulfonic acid; SEC, size exclusion chromatography; SEAP, secreted embryonic alkaline phosphatase; SPR, surface plasmon resonance

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